

Problem Set 4 revised

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LPG dependence in Nepali Households as Recorded in 2021 Nepal Census.

Summary

This report analyzes household cooking fuel use in Nepal using data from the 2021 National Population and Housing Census, with the intention of understanding the role of liquefied petroleum gas (LPG). This analysis shows that LPG has become the main cooking fuel in cities and towns, while wood remains dominant in rural areas. This pattern matters because Nepal imports all of its LPG and is therefore exposed to global supply disruptions. Recent instability around the Strait of Hormuz has contributed to shortages and price hikes that Nepali households are currently experiencing. By examining cooking fuel use across provinces, districts, and types of local governments, this report shows that dependence on LPG is highly uneven within Nepal. Urban areas benefit most from LPG, so they are also the most vulnerable when imports are disrupted. The findings highlight an important trade-off between cleaner cooking and energy security in a net importer country like Nepal.

Motivation

Over the past decade, Nepal has made progress in moving households away from traditional cooking fuels such as firewood, biogas, cow dung etc. and LPG has played a central role in this transition. However, Nepal does not produce LPG domestically and relies entirely on imports, mostly through regional supply chains connected to global energy markets. This dependence has cost household a large price along many crisis that the Nepali economy has faced, like the 2015 earthquake, India's economic blockage, Covid etc.; And now is even more visible during recent shortages and price increases linked to geopolitical tensions affecting the Strait of Hormuz. When supply routes are disrupted, Nepali households feel the effects almost immediately. Understanding how widespread LPG use is and where it is most concentrated, is important for energy planning and policy. National averages alone do not show which regions are most exposed to LPG shortages or which areas are less affected because they still rely on

domestic fuels. This report aims to fill that gap by using detailed census data to map cooking fuel use across Nepal and assess the country’s vulnerability to LPG supply shocks.

Data and Measurement

The analysis uses data from Nepal’s 2021 National Population and Housing Census, retrieved from [Government of Nepal, National Statistics Office](#), which reports the number of households using different fuels for cooking. The fuels included in the dataset are Wood, LPG, Electricity, Bio gas, Cow dung, Kerosene, and Other sources. For visualization purpose, all sources except for the two big ones i.e. Wood and LPG, have been consolidated as ‘Other sources’. The data are available at multiple levels, including province, district, and local government.

Local governments are grouped into four types, from most to least urban:

1. Metropolitan City
2. Sub-Metropolitan City
3. Municipality
4. Gaunpalika (rural municipalities)

Table 1: Percentage Distribution of Cooking Fuel by Local Level Type

Local_type	National_Share	Wood	LPG	Others
Metropolitan City	9.9%	7%	91.6%	1.5%
Sub-Metropolitan City	6.8%	26.4%	71.1%	2.5%
Municipality	50.5%	47.4%	47.9%	4.7%
Gaunpalika	32.8%	75%	18.9%	6.1%

Source: Census 2021.

Nepal is politically divided into 7 states, that span cross 77 districts and account for 753 local levels. The four local level division are based on population size, urban/rural character, and administrative needs.

Table 1 quantifies the distribution of cooking fuel use across these local government types. It shows that while metropolitan cities account for only 9.9% of households nationally, 91.6% of those households rely on LPG as their primary cooking fuel. In contrast, Gaunpalikas, which account for 32.8% of households, have only 18.9% LPG use and 75% reliance on wood. This stark difference highlights the strong association between urbanization and LPG adoption.

To capture both proportions and scale, this report combines percentage data (Table 1) with absolute household counts (Table 2 and Table 3) and visualizations (Figures 1, 2 and Figure 3).

Analysis

Cooking fuel use by province

Figure 1 shows substantial variation in cooking fuel use across Nepal's provinces, driven largely by differences in urbanization. Provinces with larger urban populations show higher reliance on LPG, while more rural provinces remain dominated by wood. Bagmati Province stands out as the most LPG-dependent region. Table 2 confirms this pattern, showing that metropolitan and sub-metropolitan areas in Bagmati account for over

430,000 LPG-using households, compared to fewer than 100,000 households using wood in those same urban areas. However, this pattern reverses within rural parts of Bagmati. In Gaunpalikas, Table 2 shows that 272,337 households use wood compared to only 72,521 using LPG, indicating that wood use remains nearly four times higher in rural areas of the same province. In contrast, provinces such as Karnali and Sudurpaschim are overwhelmingly dependent on wood. We see that in Karnali's Gaunpalikas alone, 157,434 households rely on wood compared to just 8,898 households using LPG, indicating that the vast majority of rural households still rely on traditional fuels. These differences demonstrate that national increases in LPG use are driven primarily by urban provinces rather than by uniform adoption across the country.

Differences within provinces

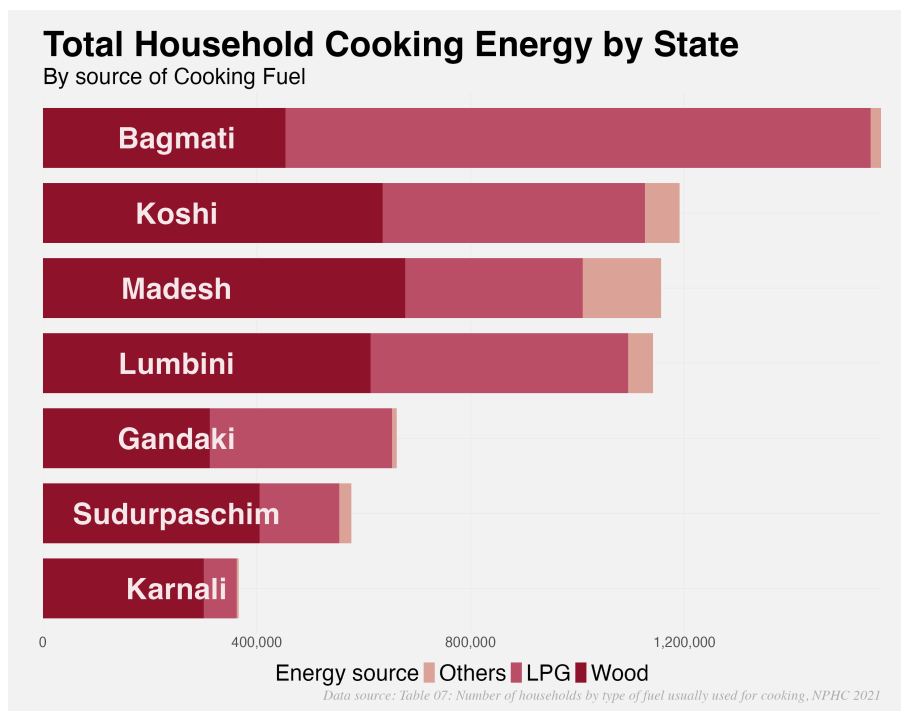


Figure 1: Cooking Energy by Province

While provincial averages reveal broad trends, Figure 3 highlights substantial variation within provinces at the district level. Table 3 indicates that 235,688 households use LPG in Kathmandu district's metropolitan areas compared to only 511 households using wood, meaning LPG use exceeds wood use by more than 400 times. In contrast, nearby districts such as Ramechhap and Sindhuli show the opposite pattern. The table shows that wood-using households exceed LPG users by more than a tenfold margin in these districts, reflecting much lower levels of urbanization and infrastructure access. Similar disparities appear in other provinces. In Jhapa District (Koshi Province), we see strong LPG use in municipalities (123,904 households), while rural Gaunpalikas continue to have tens of thousands of wood-using households. These patterns confirm that vulnerability to LPG shortages is highly localized, varying dramatically even within the same province.

Urban–rural divide

Table 1 clearly demonstrates the national urban–rural divide in cooking fuel choice, while 2 shows the share of urban and rural areas in each province based on local government types. This helps us contextualize urban-LPG and rural-wood pattern across provinces seen in 1. This pattern is consistent across provinces. For example, in Madesh Province, Table 2 shows that metropolitan areas have significantly higher LPG use, while Gaunpalikas

have 195,522 wood-using households compared to only 51,992 LPG users, meaning wood use exceeds LPG by nearly four times. Similarly, in Sudurpaschim, we see 183,206 households using wood compared to just 18,921 using LPG in Gaunpalikas, indicating extremely low LPG

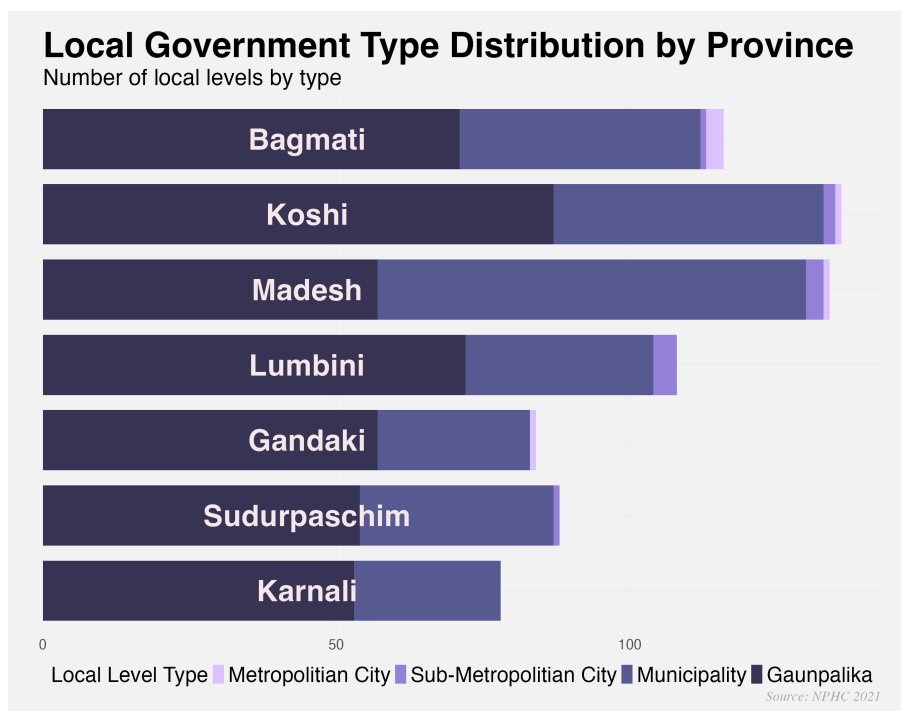


Figure 2: Local level by province

penetration. These urban areas with LPG dependence are also the most densely populated areas. As a result, although rural areas rely more heavily on traditional fuels, the majority of LPG demand, and therefore exposure to import disruptions, is concentrated in cities and large municipalities.

Discussion and Conclusion

This analysis clarifies the core trade-off in Nepal's cooking energy transition. LPG adoption has delivered cleaner and faster cooking for urban households, where over 70% of municipal households and more than 90% of metropolitan households rely on LPG. However, this same concentration means that urban provinces such as Bagmati, Koshi, and Lumbini account for the bulk of national LPG consumption, despite representing a smaller share of total local governments. Rural areas remain heavily dependent on wood, which reduces their exposure to global LPG price shocks but perpetuates health and environmental risks. For example, in Karnali and Sudurpaschim, over three-quarters of households still cook with wood, making LPG shortages less disruptive in the short run but locking households into polluting fuels. Overall, the data shows that Nepal's vulnerability to LPG supply shocks is not proportional to population alone, but rather to urban concentration. Expanding alternatives such as electric cooking or biogas in metropolitan and municipal areas where LPG use exceeds wood by margins of 5:1 or more, could substantially reduce national exposure to imported LPG while preserving the gains from cleaner cooking.

Thank you

Appendix

```
#Setup----

library(tidyverse)
library(knitr)
library(kableExtra)
library(scales)

#Themes----
mytable_theme <- function(table,
                           caption = NULL,
                           footnote_text = NULL)

table |>
  mutate(
```

```

    across(
      where(is.numeric),
      ~ ifelse(.x == 0, "-", .x)
    )
  ) |>
  kbl(
    caption = caption,
    align = c('l', rep('c', ncol(table) - 1)),
    booktabs = TRUE,
    longtable = TRUE
  ) |>
  kable_styling(
    position = "center",
    latex_options = "hold_position"
  ) |>
  row_spec(0, bold = TRUE, background = "#2E86AB", color = "#F2F0EF") |>
  column_spec(1, bold = TRUE, background = "#F2F0EF", color = "#2E86AB") |>
  footnote(
    general = footnote_text,
    footnote_as_chunk = TRUE,
    escape = FALSE,
    general_title = ""
  )
)

#Functions----
classify_local <- function(data) {
  data |>
  mutate(
    Local_type = case_when(
      str_detect(Local_level, "Gaunp") ~ "Gaunpalika",
      str_detect(Local_level, "Munici") ~ "Municipality",
      str_detect(Local_level, "Sub-Metro") ~ "Sub-Metropolitan City",
      str_detect(Local_level, "Metro") ~ "Metropolitan City",
      TRUE ~ "Other"
    )
  )
}

fuel_summary <- function(data, group_var, include_province = FALSE) {
  if (include_province) {
    data |>
    classify_local() |>
    group_by(Province, {{ group_var }}, Local_type) |>

```

```

    summarise(
      across(c(Wood, LPG, Others), sum),
      .groups = "drop"
    ) |>
    arrange(
      Province, {{ group_var }},
      match(Local_type, c("Metropolitan City", "Sub-Metropolitan City",
        ↪ "Municipality", "Gaunpalika"))
    )
  } else {
data |>
  classify_local() |>
  group_by({{ group_var }}, Local_type) |>
  summarise(
    across(c(Wood, LPG, Others), sum),
    .groups = "drop"
  ) |>
  arrange(
    {{ group_var }},
    match(Local_type, c("Metropolitan City", "Sub-Metropolitan City",
      ↪ "Municipality", "Gaunpalika"))
  )
  }
}

count_local <- function(data, group_var, include_province = FALSE) {
  if (include_province) {
    data |>
      classify_local() |>
      distinct(Province, {{ group_var }}, Local_level, Local_type) |>
      count(Province, {{ group_var }}, Local_type) |>
      pivot_wider(names_from = Local_type, values_from = n, values_fill = 0)
      ↪ |>
      select(
        Province, {{ group_var }},
        any_of(c("Gaunpalika", "Municipality", "Sub-Metropolitan City",
          ↪ "Metropolitan City"))
      ) |>
      arrange(Province, {{ group_var }})
  } else {
    data |>
      classify_local() |>

```

```

distinct({{ group_var }}, Local_level, Local_type) |>
count({{ group_var }}, Local_type) |>
pivot_wider(names_from = Local_type, values_from = n, values_fill = 0)
  ↪ |>
select(
  {{ group_var }},
  any_of(c("Gaunpalika", "Municipality", "Sub-Metropolitan City",
  ↪ "Metropolitan City"))
)
}
}

fuel_table <- function(data, caption, group_col = 1, font = 8) {
  mytable_theme(
    data,
    caption = caption,
    footnote_text = "Source: Census 2021."
  ) |>
  kable_styling(
    latex_options = c("repeat_header", "scale_down"),
    font_size = font
  ) |>
  collapse_rows(columns = group_col, valign = "top")
}

# Data----
cooking <- read.csv('Cooking_fuel_NP.csv') |>
mutate(
  Others = Electricity + Cow_dung + Bio_gas + Kerosene + Other
) |>
select(Province, District, Local_level, Total, Wood, LPG, Others )

energy_source <- c("Wood", "LPG", "Others")
energy_colors <- c(Wood = "#890620", LPG = "#b6465f", Others = "#da9f93")

#Table prep----
fuel_district <- fuel_summary(cooking, District, include_province = TRUE)
fuel_state <- fuel_summary(cooking, Province)
count_dist <- count_local(cooking, District, include_province = TRUE)
count_prov <- count_local(cooking, Province)

```



```

province_order <- cooking |>
  group_by(Province) |>
  summarise(total = sum(Total)) |>
  arrange(total) |>
  pull(Province)

```

```

energy_df_prov <- cooking |>
  select(Province, all_of(energy_source))

barplot_province <- aggregate(
  energy_df_prov[, energy_source],
  by = list(Province = energy_df_prov$Province),
  sum
) |>
  pivot_longer(
    cols = all_of(energy_source),
    names_to = "Source",
    values_to = "NumHouse"
  ) |>
  group_by(Province) |>
  mutate(total = sum(NumHouse)) |>
  ungroup() |>
  mutate(
    Province = factor(Province, levels = province_order),
    Source = fct_reorder(Source, NumHouse, sum)
  )

stacked <- barplot_province |>
  ggplot(aes(
    x = NumHouse,
    y = Province,
    fill = Source
  )) +
  geom_bar(
    width = 0.8,
    stat = "identity",
    position = "stack",
    alpha = 0.95
  ) +
  geom_text(
    data = distinct(barplot_province, Province, total),
    aes(

```

```

    x = 250000,
    y = Province,
    label = Province
  ),
  inherit.aes = FALSE,
  color = "#f6e8ea",
  size = 15,
  fontface = "bold"
) +
scale_fill_manual(values = energy_colors) +
scale_x_continuous(
  labels = scales::comma,
  expand = expansion(mult = c(0, 0))
) +
labs(
  title = "Total Household Cooking Energy by State",
  subtitle = "By source of Cooking Fuel",
  fill = "Energy source",
  caption = "Data source: Table 07: Number of households by type of fuel
  ↪ usually used for cooking, NPHC 2021"
) +
theme_minimal() +
theme(
  plot.title = element_text(size = 50, hjust = 0, face = "bold", family =
  ↪ "sans"),
  plot.title.position = "plot",
  plot.subtitle = element_text(hjust = 0, size = 32, family = "sans"),
  plot.background = element_rect(fill = "#f2f2f2", color = NA),
  panel.grid.minor = element_blank(),
  axis.title = element_blank(),
  axis.text.y = element_blank(),
  axis.text.x = element_text(size = 20),
  axis.ticks.y = element_blank(),
  plot.caption.position = "plot",
  plot.caption = element_text(hjust = 1, size = 20, face = "italic", family
  ↪ = "times", color = 'darkgray'),
  legend.position = "bottom",
  legend.text = element_text(size = 32),
  legend.title = element_text(size = 32),
  plot.margin = margin(t = 30, r = 50, b = 20, l = 50, unit = "pt")
)

ggsave("stacked.png", plot = stacked, width = 18, height = 14, dpi = 150)

```

```

energy_df_dist <- cooking |>
  select(Province, District, all_of(energy_source))

barplot_district <- aggregate(
  energy_df_dist[, energy_source],
  by = list(Province = energy_df_dist$Province, District =
    ↪ energy_df_dist$District),
  sum
) |>
pivot_longer(
  cols = all_of(energy_source),
  names_to = "Source",
  values_to = "NumHouse"
) |>
group_by(District) |>
mutate(total = sum(NumHouse)) |>
ungroup() |>
mutate(
  Province = factor(Province, levels = province_order),
  District = fct_reorder(District, total),
  Source = fct_reorder(Source, NumHouse, sum),
  label_hjust = 1.20,
  label_color = "#f6e8ea"
)

stacked_districts <- barplot_district |>
  ggplot(aes(x = NumHouse, y = District, fill = Source)) +
  geom_bar(stat = "identity", position = "stack", alpha = 0.95) +
  geom_label(
    data = distinct(barplot_district, District, Province, total,
      label_hjust, label_color),
    aes(
      x = 350000,
      y = District,
      label = District),
    size = 6,
    fill = '#FAECEC',
    color = 'black',
    style = 'bold',
    inherit.aes = FALSE
  ) +

```

```

scale_color_identity() +
scale_fill_manual(values = energy_colors) +
scale_x_continuous(
  labels = scales::comma,
  expand = expansion(mult = c(0, 0.35))
) +
facet_wrap(~ Province, scales = "free_y", ncol = 2) +
labs(
  title = "Total Household Cooking Energy by District",
  subtitle = "By source of Cooking Fuel",
  fill = "Energy Source",
  caption = "Data source: Table 07: Number of households by type of fuel
    ↪ usually used for cooking, NPHC 2021"
) +
theme_minimal() +
theme(
  plot.title = element_text(size = 42, hjust = 0, face = "bold", family =
    ↪ "sans"),
  plot.title.position = "plot",
  plot.subtitle = element_text(hjust = 0, size = 30, family = "sans"),
  plot.background = element_rect(fill = "#F2F0EF", color = NA),
  panel.grid.minor = element_blank(),
  axis.title = element_blank(),
  axis.text.y = element_blank(),
  axis.text.x = element_text(size = 18),
  axis.ticks.y = element_blank(),
  plot.caption.position = "plot",
  plot.caption = element_text(hjust = 1, face = "italic", family = "sans",
    ↪ size = 18, color = 'darkgray'),
  strip.text = element_text(hjust = 0.47, size = 22, face = "bold"),
  legend.position = "top",
  legend.text = element_text(size = 20),
  legend.title = element_text(size = 19),
  plot.margin = margin(t = 30, r = 20, b = 20, l = 20, unit = "pt")
)

ggsave("stacked_districts.png", plot = stacked_districts, width = 20, height
  ↪ = 26, dpi = 150)

```

```

fuel_table(fuel_state, "Cooking Fuel by Local Level Type per State", font =
  ↪ 7)

```

Table 2: Cooking Fuel by Local Level Type per State

Province	Local_type	Wood	LPG	Others
Bagmati	Metropolitan City	13705	392880	6131
	Sub-Metropolitan City	8122	37806	638
	Municipality	159823	590988	9880
	Gaunpalika	272337	72521	3086
Gandaki	Metropolitan City	11893	127571	995
	Municipality	133272	158515	4986
	Gaunpalika	166760	54931	2709
Karnali	Municipality	143528	52829	2092
	Gaunpalika	157434	8898	1256
Koshi	Metropolitan City	9270	45657	1992
	Sub-Metropolitan City	12827	78541	1378
	Municipality	294795	287077	23957
	Gaunpalika	318780	79022	37459
Lumbini	Sub-Metropolitan City	52433	124493	3983
	Municipality	215319	230168	19554
	Gaunpalika	344946	127087	23362
Madesh	Metropolitan City	11020	35580	514
	Sub-Metropolitan City	32358	52266	3560
	Municipality	438832	192154	84055
	Gaunpalika	195522	51992	58530
Sudurpaschim	Sub-Metropolitan City	13952	29261	1566
	Municipality	208182	100837	13464
	Gaunpalika	183206	18921	7383

Source: Census 2021.

```

fuel_percent <- fuel_summary(cooking, Province) |>
  group_by(Local_type) |>
  summarise(
    across(c(Wood, LPG, Others), sum),
    Total = sum(Wood, LPG, Others),
    .groups = "drop"
  ) |>
  mutate(
    National_total = sum(Total),
    National_Share = round((Total / National_total) * 100, 1),
    across(c(Wood, LPG, Others),
      ~ round(.x / Total * 100, 1)),
    across(

```

```

    -Local_type,
    ~ paste0(.x, "%")
  )
) |>
select(
  Local_type, National_Share,
  Wood, LPG, Others
) |>
arrange(match(Local_type,
              c("Metropolitan City",
                "Sub-Metropolitan City",
                "Municipality",
                "Gaunpalika")))

```

```

fuel_table(fuel_percent, "Percentage Distribution of Cooking Fuel by Local
↪ Level Type")

```

```

fuel_table(fuel_district, "Cooking Fuel by Local Level Type per District",
  group_col = c(1,2),
  font = 7)

```

Table 3: Cooking Fuel by Local Level Type per District

Province	District	Local_type	Wood	LPG	Others
	Bhaktapur	Municipality	7969	99180	1257
		Metropolitan City	9824	84961	1806
	Chitawan	Municipality	19272	54681	2438
		Gaunpalika	3718	2407	60
	Dhading	Municipality	11795	11340	502
		Gaunpalika	39950	18962	1073
	Dolakha	Municipality	9353	4866	189
		Gaunpalika	31957	2936	192
	Kathmandu	Metropolitan City	511	235688	2767
		Municipality	9764	291243	2919
	Kavrepalanchok	Municipality	20537	40783	959
		Gaunpalika	24762	4046	243
	Lalitpur	Metropolitan City	3370	72231	1558
		Municipality	4820	50648	683
		Gaunpalika	5453	1309	58

Table 3: Cooking Fuel by Local Level Type per District (*continued*)

Province	District	Local_type	Wood	LPG	Others
Gandaki	Makwanpur	Sub-Metropolitan City	8122	37806	638
		Municipality	6095	3582	72
		Gaunpalika	35327	13654	324
	Nuwakot	Municipality	12992	11247	217
		Gaunpalika	32760	11118	312
	Ramechhap	Municipality	14019	3549	50
		Gaunpalika	26631	2006	211
	Rasuwa	Gaunpalika	7565	3517	49
		Municipality	20371	13428	446
	Sindhuli	Gaunpalika	29468	5354	250
		Municipality	22836	6441	148
	Sindhupalchok	Gaunpalika	34746	7212	314
		Municipality	25949	11862	140
	Baglung	Gaunpalika	24098	1968	79
		Municipality	11276	13983	801
	Gorkha	Gaunpalika	34537	10477	655
		Metropolitan City	11893	127571	995
	Kaski	Gaunpalika	13889	5969	99
		Municipality	11875	17628	752
	Lamjung	Gaunpalika	11088	2571	160
		Gaunpalika	1058	454	35
	Manang	Gaunpalika	1209	1951	446
	Mustang	Municipality	4880	4412	44
		Gaunpalika	16255	2973	202
	Nawalparasi (East)	Municipality	19233	51675	1372
		Gaunpalika	13885	7290	395
	Parbat	Municipality	10942	5694	63
		Gaunpalika	15087	4249	77
	Syangja	Municipality	26298	18552	810
		Gaunpalika	18995	4091	177
	Tanahu	Municipality	22819	34709	1004
		Gaunpalika	16659	12938	384
Lumbini	Dailekh	Municipality	23275	3242	287
		Gaunpalika	25616	1973	201
	Dolpa	Municipality	4274	428	247
		Gaunpalika	3804	129	498
	Humla	Gaunpalika	11035	133	36
		Municipality	19480	2441	142
	Kailash	Municipality	19480	2441	142
		Gaunpalika	19480	2441	142

Table 3: Cooking Fuel by Local Level Type per District (*continued*)

Province	District	Local_type	Wood	LPG	Others
	Jajarkot	Gaunpalika	14490	790	110
		Municipality	3541	1789	37
	Jumla	Gaunpalika	18483	519	53
		Municipality	9039	1272	138
	Kalikot	Gaunpalika	15448	778	95
		Municipality	4007	802	143
	Mugu	Gaunpalika	7125	330	23
		Municipality	19697	3243	143
	Rukum (West)	Gaunpalika	13334	834	39
		Municipality	20199	3563	62
	Salyan	Gaunpalika	28846	1861	141
		Municipality	40016	36049	893
	Surkhet	Gaunpalika	19253	1551	60
		Municipality	12008	1965	82
	Bhojpur	Gaunpalika	23600	871	54
		Municipality	13160	6659	74
	Dhankuta	Gaunpalika	14450	3131	142
		Municipality	31907	10040	662
	Ilam	Gaunpalika	23272	4185	435
		Municipality	45179	123904	5565
	Jhapa	Gaunpalika	34375	29617	6379
		Municipality	14305	2297	200
	Khotang	Gaunpalika	23725	1054	139
		Metropolitan City	9270	45657	1992
	Morang	Municipality	52153	73194	10323
		Gaunpalika	41632	26511	11428
	Okhaldhunga	Municipality	6006	1036	84
		Gaunpalika	25279	1688	193
	Panchthar	Municipality	8525	3705	106
		Gaunpalika	27929	1978	194
	Sankhuwasabha	Municipality	21857	5189	251
		Gaunpalika	11065	718	38
	Solukhumbu	Municipality	4955	1777	12
		Gaunpalika	16830	2168	497
	Sunsari	Sub-Metropolitan City	12827	78541	1378
		Municipality	32550	30763	5217
		Gaunpalika	28910	4463	17758
		Municipality	4560	2285	53

Table 3: Cooking Fuel by Local Level Type per District (*continued*)

Province	District	Local_type	Wood	LPG	Others
Lumbini	Tapijung	Gaunpalika	20070	721	87
		Municipality	6258	2288	85
	Terhathum	Gaunpalika	12450	728	36
		Municipality	41372	21975	1243
	Udayapur	Gaunpalika	15193	1189	79
		Municipality	24213	4989	78
	Arghakhanchi	Gaunpalika	18569	574	26
		Sub-Metropolitan City	7200	26899	466
	Banke	Municipality	5617	18030	536
		Gaunpalika	45242	23340	1904
	Bardiya	Municipality	55980	25142	4440
		Gaunpalika	11546	7389	1788
	Dang	Sub-Metropolitan City	42583	49975	3221
		Municipality	5363	7121	974
		Gaunpalika	37924	13607	1498
	Gulmi	Municipality	11581	4419	114
		Gaunpalika	45439	4276	271
	Kapilbastu	Municipality	47153	36670	6994
		Gaunpalika	15403	9452	6189
	Nawalparasi (West)	Municipality	15144	32632	1280
		Gaunpalika	19024	13931	698
	Palpa	Municipality	10495	15221	429
		Gaunpalika	32248	6181	417
	Pyuthan	Municipality	16037	3345	71
		Gaunpalika	33807	2757	178
	Rolpa	Municipality	6959	1557	229
		Gaunpalika	40483	2697	281
	Rukum (East)	Gaunpalika	11966	830	82
	Rupandehi	Sub-Metropolitan City	2650	47619	296
		Municipality	16777	81042	4409
		Gaunpalika	33295	42053	10030
	Bara	Sub-Metropolitan City	23311	23009	1455
		Municipality	28805	11624	3015
		Gaunpalika	26499	10405	3056
	Dhanusa	Sub-Metropolitan City	9047	29257	2105
		Municipality	62055	28312	15356
		Gaunpalika	16497	8744	5718
		Municipality	64485	29929	9128

Table 3: Cooking Fuel by Local Level Type per District (*continued*)

Province	District	Local_type	Wood	LPG	Others
Madhesh	Mahottari	Gaunpalika	23738	7290	3316
	Parsa	Metropolitan City	11020	35580	514
		Municipality	14522	5427	716
		Gaunpalika	34599	9389	1246
	Rautahat	Municipality	79822	34090	14278
		Gaunpalika	5747	1779	1309
	Saptari	Municipality	53831	15342	23317
		Gaunpalika	27164	3740	23422
	Sarlahi	Municipality	72793	38769	5261
		Gaunpalika	36927	6120	4954
	Siraha	Municipality	62519	28661	12984
		Gaunpalika	24351	4525	15509
	Achham	Municipality	20288	2047	179
		Gaunpalika	26062	881	110
Sudurpaschim	Baitadi	Municipality	23545	1702	74
		Gaunpalika	23473	579	34
	Bajhang	Municipality	9755	1539	80
		Gaunpalika	25566	1014	71
	Bajura	Municipality	14548	1642	262
		Gaunpalika	11203	358	28
	Dadeldhura	Municipality	10563	2959	94
		Gaunpalika	16787	729	40
	Darchula	Municipality	6783	3667	33
		Gaunpalika	17074	776	48
	Doti	Municipality	12119	3234	48
		Gaunpalika	28465	1079	195
	Kailali	Sub-Metropolitan City	13952	29261	1566
		Municipality	60414	39741	5446
		Gaunpalika	27291	11775	6426
	Kanchanpur	Municipality	50167	44306	7248
		Gaunpalika	7285	1730	431

Source: Census 2021.

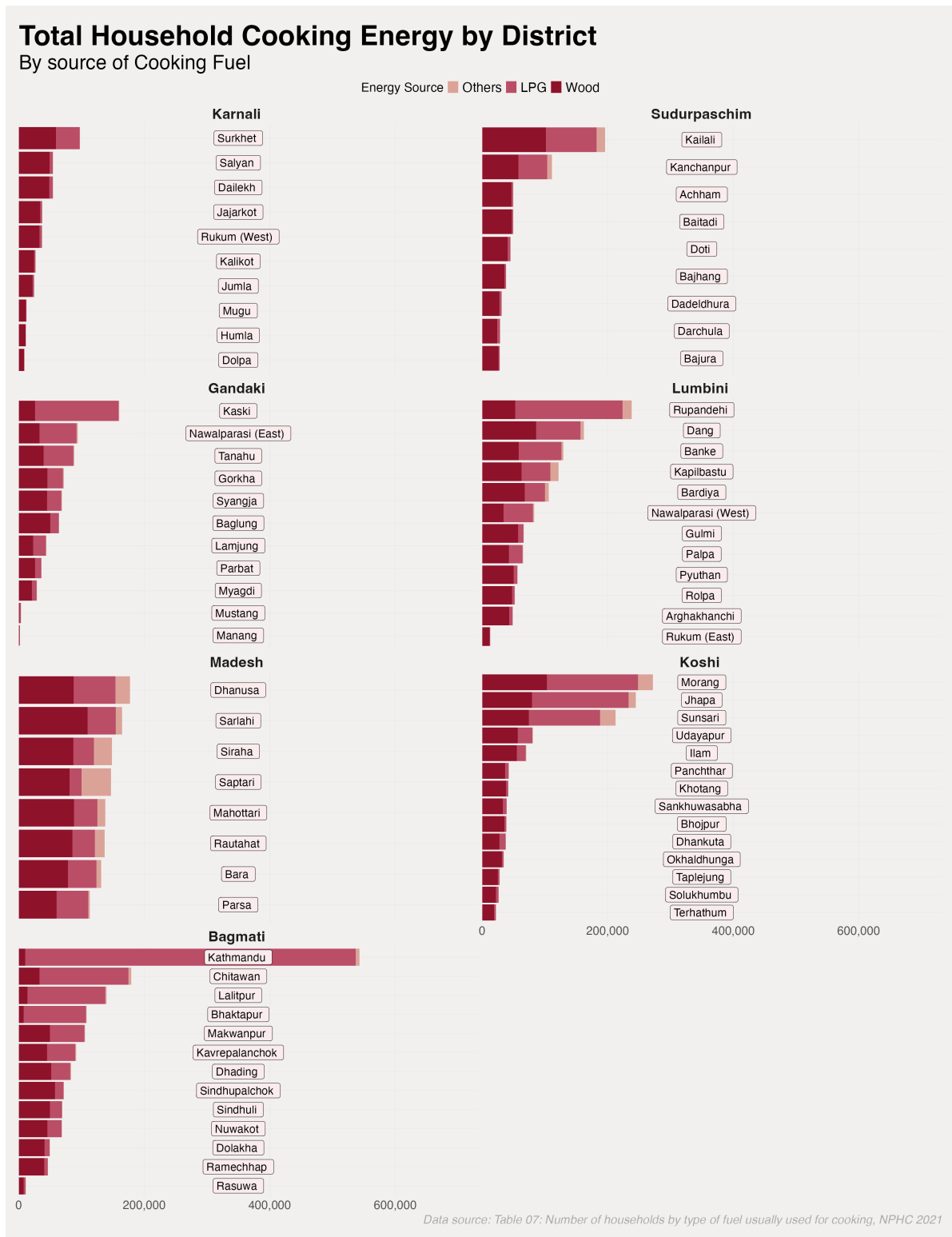


Figure 3: Cooking Energy by District